

Methods: detection of coordinated calcium events

Input. A *coordinated event* is a group of calcium-event onsets in several regions of interest (ROIs, one cell each) within about a second; a detector reports each one it finds as a *call*, an interval of time. The input is the imaging pipeline's per-ROI event table: each event's onset, its half-rise time (t_{50}), on a 0.1 s frame grid. The pipeline exports two event sets, *fast* and *slow*, assigned upstream; a third, *combined*, stream is the time-sorted union of each ROI's fast and slow onsets, near-coincident pairs kept. Each recording is divided into *windows*, its baseline and treatment periods. The data are 66 recordings (2,109 ROIs, 36 mice) whose first treatment was TTX or senktide, in four groups: diestrus females (DI, 17 recordings), ovariectomized females (OVX, 17), males (MALE, 13) and orchidectomized males (ORX, 19). Group is nested in imaging day.

Event floor. A call needs at least the *event floor* of ROIs co-active within 2 s: the smallest count chance reaches at most once per hour, and never fewer than 3 ROIs. Chance comes from 1,000 draws that shift each ROI's onset train as a whole by its own offset, uniform in $[-20\text{ s}, 20\text{ s}]$, keeping each ROI's rate and intervals (after Pipa et al., 2008, but dropping rather than wrapping onsets pushed past the window's edge). The null keeps coordinated onsets, so coordination-rich windows get a higher, conservative floor. Each window of each stream, and each simulated recording, has its own floor; treatment windows are also scored under their baseline floor.

Parameters for simulation. Detectors were tuned and scored on simulated recordings with known (*planted*) coordinated events, whose statistics were measured on baseline windows and pooled over groups. *Spacing*: coordinated events were located, without the detectors compared here, as runs of the co-active count at or above the floor (peaks under 2 s apart merged). Median gaps between neighbors were 41.3 s (fast), 25.1 s (slow) and 24.8 s (combined), at 9.7, 22.0 and 25.3 events per hour of baseline; events cluster, so the median gap is short. *Participation*: median participants per cluster of at least 4 co-active ROIs over median ROI count (fast 0.20, slow 0.38, combined 0.25) set the middle of three planted levels; the outer two were chosen. *Background*: quiet and busy per-ROI event rates are the 25th and 75th percentiles across recordings, coordinated onsets removed, varied across ROIs and over time by Gamma factors fitted to the data. *Timing spread*: the standard deviation (SD) of onsets within an event (fast 0.105 s, slow 0.131 s, combined 0.150 s) came from the half-width of the cross-ROI onset correlogram (Perkel et al., 1967), calibrated on simulations of known spread.

Simulated recordings. Each has 32 ROIs (the measured median) and lasts 45 min. It plants the stream's measured events per 45 min (fast 7, slow 17, combined 19) over the three levels, laid end to end at gaps resampled from the pooled gaps and placed at a random time, plus six *decoys* (bursts built like middle-level events but not scored as events), on a quiet or busy background. Each seed also yields a *no-coordination recording* (quiet, nothing planted) and an *elevated-rate recording* (a 300 s stretch at the 99th percentile of background rate). A variant spaced like ORX checks the rankings. Separate seeds serve *selection*, *held-out* confirmation and *final* scoring; fast gets twice as many, having fewer events.

Scoring. A call matches a planted event lying in its span or within 2.5 s of it, one to one, closest pairs first. A planted event under its recording's floor, and any call matched to it, leave the score. F1, the harmonic mean of precision and recall, is pooled per background and averaged over the two; calls on decoys count against precision. Calls holding two or more scored events are counted as merges. Every setting must meet budgets set at $1.6\times$ the rates of CoactDetect's released settings: calls per hour on the no-coordination recording (fast 7, slow 1, combined 10), 1 call per minute in the elevated stretch, and a quiet-to-busy precision difference of at most 0.10.

CoactDetect. $S(t)$ counts the ROIs with an onset in $(t - w, t]$, w being the count interval. Its mean and variance under a null that circularly shifts each ROI's onsets within a surrounding *context*, optionally excluding a guard band around t , are computed exactly; t is called when S reaches the floor and $z = (S - \text{mean})/\text{SD}$ passes a cutoff α stated as a normal tail, a z threshold rather than a false-alarm probability. Calls closer than a merge gap are joined. The design resembles cell-averaging constant-false-alarm-rate radar detection (Finn & Johnson, 1968), reached independently. Settings were searched in this sliding form (fast's released setting uses fixed 2 s bins) by coordinate search from the released settings: α down to 6×10^{-16} ($z \approx 8$), contexts 20–120 s, guard at most a quarter of the context. The floor is always computed at 2 s, whatever w . A candidate was adoptable if its held-out F1 gain had a 95% bootstrap interval above zero, its best values lay inside the tested range, and it met every budget; a best value at a cap, or one that switches a setting off, was reported, not adopted.

Chorus. This learned detector reads a binary ROI $\times$ frame onset raster. A four-layer dilated convolutional stack (Yu & Koltun, 2016) shared by all ROIs filters each ROI; each channel is standardized over the whole input and passed through a sigmoid as a per-ROI vote. Votes are pooled across ROIs (mean, SD, mean of the top four), so any number of ROIs in any order is accepted (Qi et al., 2017; Zaheer et al., 2017), and a dilated head scores each frame; frames over a threshold chosen on validation recordings become calls, runs within 2 s merged. Standardization spans 409.6 s excerpts in training and the whole window at inference. A participation variant sums the votes into a bounded count, takes the floor as input and learns which ROIs took part. Training was per stream (Adam, Kingma & Ba, 2015; learning rate 0.01; 900 steps; class-weighted binary cross-entropy), with extra events planted at the floor and one ROI either side, sub-floor events labelled negative. Of five initializations, the best on selection recordings within the no-coordination budget was kept (the best overall if none met it).

Benchmark. Each tuned or trained detector was compared with CoactDetect on the final seeds as a paired F1 difference with a 95% bootstrap interval over seeds (2,000 resamples; Efron, 1979). A detector more than 0.01 F1 behind the best on at least two streams is dropped from further tuning.

Output. Every call is measured the same way, whatever detector made it. Onsets within ± 1 s (fast) or ± 5 s (slow, combined) of the call's center, widened to its span, are split where consecutive onsets are over 0.5 s (fast) or 2.5 s (slow, combined) apart; the group with most distinct ROIs is the event's *core*. These lengths were chosen, not measured. The event time is the core's first onset; *width* is the core's first-to-last onset span, floored at one frame; *amplitude* is core ROIs divided by width (ROIs s^{-1}), undefined for one ROI. Amplitude measures packing, not fluorescence, is dominated by width, and has a provisional name. The core's ROI count has been checked against planted participants on simulated data only, where it over-counts on combined; width and amplitude are unchecked.

References

- Efron B (1979). Bootstrap methods: another look at the jackknife. *Annals of Statistics* 7(1):1–26. doi:10.1214/aos/1176344552
- Finn HM, Johnson RS (1968). Adaptive detection mode with threshold control as a function of spatially sampled clutter-level estimates. *RCA Review* 29(3):414–464.
- Kingma DP, Ba J (2015). Adam: a method for stochastic optimization. *International Conference on Learning Representations*. arXiv:1412.6980
- Perkel DH, Gerstein GL, Moore GP (1967). Neuronal spike trains and stochastic point processes. II. Simultaneous spike trains. *Biophysical Journal* 7(4):419–440. doi:10.1016/S0006-3495(67)86597-4
- Pipa G, Wheeler DW, Singer W, Nikolić D (2008). NeuroXidence: reliable and efficient analysis of an excess or deficiency of joint-spike events. *Journal of Computational Neuroscience* 25(1):64–88. doi:10.1007/s10827-007-0065-3
- Qi CR, Su H, Mo K, Guibas LJ (2017). PointNet: deep learning on point sets for 3D classification and segmentation. *IEEE Conference on Computer Vision and Pattern Recognition*, 652–660. doi:10.1109/CVPR.2017.16
- Yu F, Koltun V (2016). Multi-scale context aggregation by dilated convolutions. *International Conference on Learning Representations*. arXiv:1511.07122
- Zaheer M, Kottur S, Ravanbakhsh S, Póczos B, Salakhutdinov R, Smola AJ (2017). Deep Sets. *Advances in Neural Information Processing Systems* 30. arXiv:1703.06114